

Scalability&FuturePlan

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TeamID	SWTID-2026-9346
ProjectName	Voice--Based--Concept--Understanding--Analyzer
MaximumMarks	1 Mark

Scalability&FuturePlan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address
1	Supports a limited number of concurrent users.	Performance may decrease when multiple users access the system simultaneously.	High
2	Local file-based storage for reports and user sessions.	Limited scalability and difficult long-term data management.	Medium
3	AI processing time depends on audio length and model execution.	Longer processing time for large audio files may affect user experience.	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade/Solution
1	User Load	Supports a small number of users.	Deploy the application on cloud infrastructure with load balancing to support multiple concurrent users.
2	Data Storage	Stores reports and session data locally.	Migrate to a cloud database such as PostgreSQL or Firebase for scalable and secure data management.
3	Performance	AI processing is sequential.	Optimize AI models, introduce asynchronous processing, and implement caching for faster response times.
4	Security	Basic authentication and file validation.	Add role-based authentication, encrypted data storage, secure API integration, and audit logging.

Future Roadmap:

Phase	Planned Feature/ Enhancement	Target Timeline	Expected Impact
Phase 2	Complete AI integration, improve semantic analysis accuracy, and	Next 2–3 Months	Improved accuracy and accessibility for diverse users.

support multiple languages.

Phase3	Develop a mobile-friendly version, user dashboard, and cloud database integration.	Next4–6Months	Better user experience, scalability, and centralized data management.
Phase4	Real-time voice evaluation, personalized learning recommendations, analytics dashboard, and deployment on cloud platforms.	Next6–12Months	Enterprise-ready solution with enhanced performance, intelligent feedback, and large-scale usability.